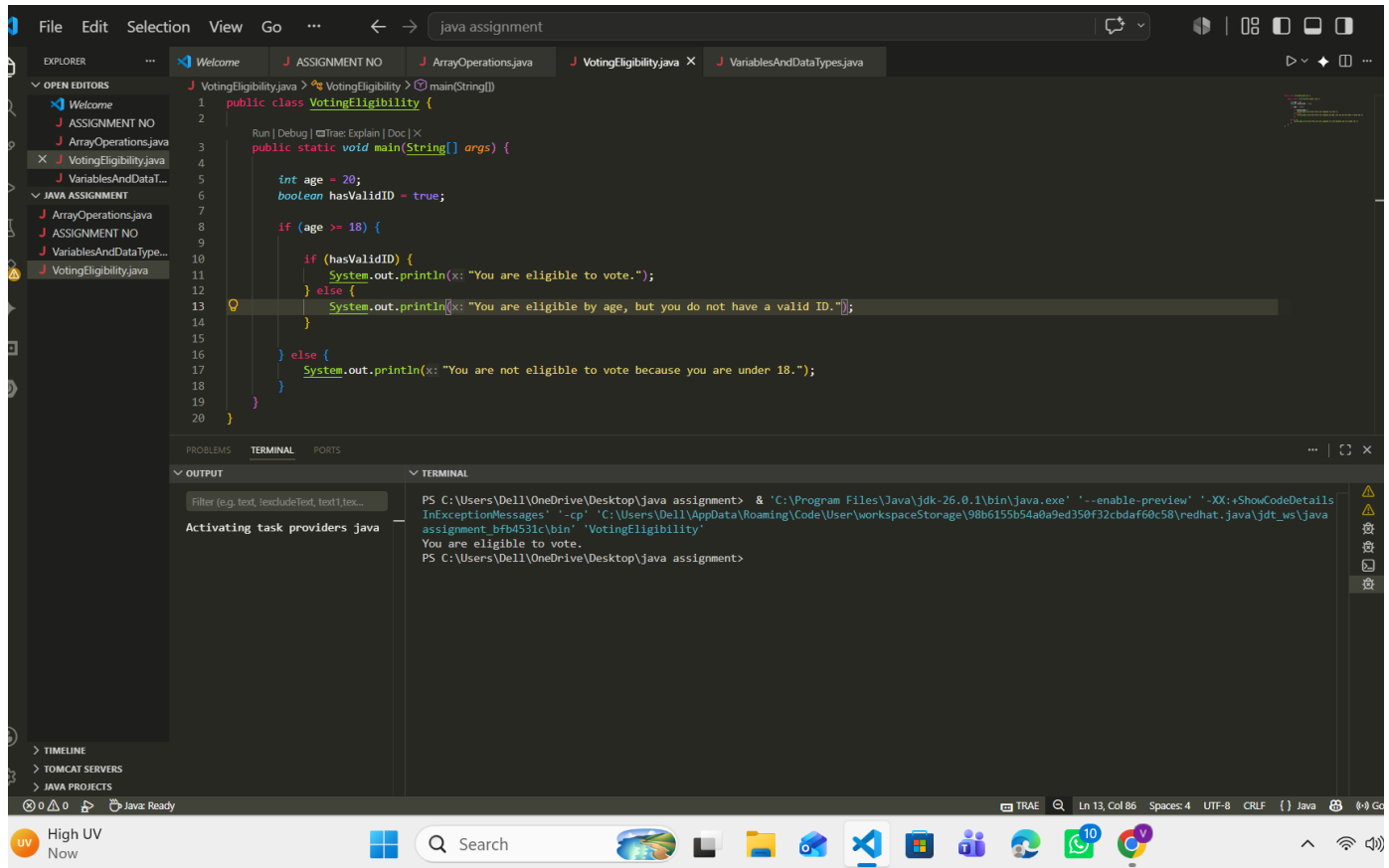


Question No. 1 – Variables and Data Types



The screenshot shows an IDE with a Java file named `VotingEligibility.java`. The code defines a class `VotingEligibility` with a `main` method that checks if a person is eligible to vote based on their age and whether they have a valid ID.

```
1 public class VotingEligibility {
2     main(String[] args) {
3
4         int age = 20;
5         boolean hasValidID = true;
6
7         if (age >= 18) {
8
9             if (hasValidID) {
10                 System.out.println(x: "You are eligible to vote.");
11             } else {
12                 System.out.println(x: "You are eligible by age, but you do not have a valid ID.");
13             }
14         } else {
15             System.out.println(x: "You are not eligible to vote because you are under 18.");
16         }
17     }
18 }
19
20 }
```

The terminal output shows the command to run the program and the resulting output:

```
PS C:\Users\Dell\OneDrive\Desktop\java assignment> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-enable-preview' '-XX:+ShowCodeDetails InExceptionMessages' '-cp' 'C:\Users\Dell\AppData\Roaming\Code\User\workspaceStorage\98b6158b54a0a9ed350f32cbdaf60c58\redhat.java\jdt_ws\java assignment_bfb4531c\bin' 'VotingEligibility'
You are eligible to vote.
PS C:\Users\Dell\OneDrive\Desktop\java assignment>
```

Question No. 2 – Array Operations

The screenshot shows a Java IDE with a file named `ArrayOperations.java` open. The code defines a class `ArrayOperations` with a `main` method that performs various operations on an array of integers: `[10, 20, 30, 40, 50]`. The operations include calculating the sum, finding the largest and smallest elements, and calculating the average. The output of the program is displayed in the `OUTPUT` pane at the bottom.

```
1 public class ArrayOperations {
2     public static void main(String[] args) {
3         // Create and initialize an array
4         int[] numbers = {10, 20, 30, 40, 50};
5
6         // Display array elements
7         System.out.println("Array elements: ");
8         for (int i = 0; i < numbers.length; i++) {
9             System.out.print(numbers[i] + " ");
10        }
11        System.out.println();
12
13        // Calculate sum, largest, and smallest
14        int sum = 0;
15        int largest = numbers[0];
16        int smallest = numbers[0];
17        for (int i = 0; i < numbers.length; i++) {
18            sum += numbers[i];
19            if (numbers[i] > largest) {
20                largest = numbers[i];
21            }
22            if (numbers[i] < smallest) {
23                smallest = numbers[i];
24            }
25        }
26
27        // Calculate average
28        double average = (double) sum / numbers.length;
29
30        // Display results
31        System.out.println("Sum of array elements: " + sum);
32        System.out.println("Average of array elements: " + average);
33        System.out.println("Largest element: " + largest);
34        System.out.println("Smallest element: " + smallest);
35    }
36 }
```

OUTPUT

```
PS C:\Users\Devl\OneDrive\Desktop\java assignment> java -cp . ArrayOperations
Array elements: 10 20 30 40 50
Sum of array elements: 150
Average of array elements: 30.0
Largest element: 50
Smallest element: 10
PS C:\Users\Devl\OneDrive\Desktop\java assignment>
```

Question No. 3 – Voting Eligibility Using If-Else and Nested If-Else

The screenshot displays an IDE with a Java project named 'java assignment3'. The main editor shows the file 'ArrayOperations.java' with the following code:

```
1 public class ArrayOperations {  
2  
3     Run [Debug] (Ctrl+F5) | Run [Run] (F5)  
4     public static void main(String[] args) {  
5         // Create and initialize an array  
6         int[] numbers = {15, 25, 35, 45, 55};  
7  
8         int sum = 0;  
9         int largest = numbers[0];  
10        int smallest = numbers[0];  
11        // Display array elements  
12        System.out.println("Array elements: ");  
13  
14        // Calculate sum, largest, and smallest  
15        for (int i = 0; i < numbers.length; i++) {  
16            System.out.print(numbers[i]);  
17  
18            if (i < numbers.length - 1) {  
19                System.out.print(", ");  
20            }  
21  
22            sum += numbers[i];  
23  
24            if (numbers[i] > largest) {  
25                largest = numbers[i];  
26            }  
27            if (numbers[i] < smallest) {  
28                smallest = numbers[i];  
29            }  
30        }  
31  
32        // Calculate average  
33        double average = (double) sum / numbers.length;  
34  
35        // Display results  
36        System.out.println();  
37        System.out.println("Sum of array elements: " + sum);  
38        System.out.println("Average of array elements: " + average);  
39        System.out.println("Largest element: " + largest);  
40        System.out.println("Smallest element: " + smallest);  
41    }  
}
```

The bottom panel shows the 'TERMINAL' window with the following output:

```
Script: 'VariablesAndDataTypes'  
==== Variables and Data Types ====  
Integer: 1: 25  
Integer: 2: 30  
Double: 1: 15.5  
Double: 2: 4.5  
Character: A  
Boolean: true  
String: Walidwadi Gassari  
  
==== Arithmetic Operations ====  
Addition (int): 30  
Subtraction (int): 10  
Multiplication (int): 200  
Addition (double): 20.8  
  
==== String Concatenation ====  
Full Name: Walidwadi Gassari  
PS C:\Users\Delil\OneDrive\Desktop\java assignment3 >
```

The terminal window also shows a message: 'Activating task providers Java'.